H.W 6

Q5

$P(C_1|x) = \frac{3}{4}$

$P(C_2|x) = \frac{1}{4}$

$$L = \begin{array}{c} \\ \\ \text{True class} \end{array} \begin{array}{c} \text{Predicted class} \\ \begin{bmatrix} 0 & 1 \\ 5 & 0 \end{bmatrix} \end{array}$$

(a) under minimum misclassification rate rule we assign to the largest posterior value so C_1.

(b) $R(C_1|x) = L_{11} P(C_1|x) + L_{21} P(C_2|x)$

$= 5 \cdot \frac{1}{4} = \frac{5}{4}$

This is the weighted sum of losses

i.e $\underbrace{L_{11}}_{\text{loss if we predict } C_1 \text{ given it is } C_1} \times P(C_1|x) + \underbrace{L_{21}}_{\text{Loss if we predicted } C_1 \text{ given it is actually } C_2} P(C_2|x)$

L_{AB} ← truth, ↑ Predicted

(c) $R(C_2|x) = L_{22} P(C_2|x) + L_{12} P(C_1|x)$

$= 3/4$

(d) C_2 as it has the lowest conditional expected loss

(e) even though C_1 has a greater probability of being true i.e $P(C_1|x) > P(C_2|x)$ the loss on misclassifing C_1 is greater hence the minimum expected loss picks C_2 as the misclassification loss is lower.

Q6

$$P(x|C_1) = N(x|0,1)$$
$$P(x|C_2) = N(x|2,1).$$
$$P(C_1) = P(C_2) = 1/2$$

(a) $$P(x|C_1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

$$P(x|C_2) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-2)^2}{2}}$$

$$\boxed{N(x|0,1)}$$

variance

mean

variable value

(b) $$P(C_k|x) = \frac{P(x|C_k)P(C_k)}{P(x)}$$

Total probability

$$P(x) = P(x|C_1)P(C_1) + P(x|C_2)P(C_2)$$

(c) & (d) $P(x|C_1) = P(x|C_2)$ $\left(\because P(C_1) = P(C_2) \right.$ $\Rightarrow$ decision boundary when $P(x|C_1) = P(x|C_2)$

$$\Rightarrow \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} = \frac{e^{-\frac{(x-2)^2}{2}}}{\sqrt{2\pi}}$$

$$\Rightarrow -\frac{x^2}{2} = -\frac{(x-2)^2}{2}$$

$$\Rightarrow x^2 = (x-2)^2 \leftarrow \text{decision boundary}$$

$$\Rightarrow \underline{x^2 = x^2 + 4 - 4x}$$

$$\Rightarrow x = 1$$

<u>Reason for decision boundary</u> :

We know we choose C_1 if $P(C_1|x) > P(C_2|x)$
decision boundary is where we cannot classify
and that is when $P(C_1|x) = P(C_2|x)$

(e)

	$x = 0$	$x = 1$	$x = 3$
$P(x\|C_1)$	$\boxed{\frac{1}{\sqrt{2\pi}} \approx 0.4}$ C_1	$\frac{1}{\sqrt{2\pi}} e^{-1/2} \approx 0.25$	$\frac{e^{-9/2}}{\sqrt{2\pi}}$
$P(x\|C_2)$	$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}} \approx 0.25$	$\frac{1}{\sqrt{2\pi}} e^{-1/2}$	$\frac{e^{-1/2}}{\sqrt{2\pi}}$ C_2

Boundary

Q7 $P(C_1|x) = \frac{3}{4}$

$P(C_2|x) = \frac{1}{4}$

$$L = \begin{bmatrix} 0 & 1 \\ 5 & 0 \end{bmatrix}$$

Same as Question 5

Q8 Same as Question 6

Q9

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\mu_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mu_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$P(C_1) = P(C_2) = 1/2$$

(a) General formula Gaussian classes with common covariance matrix Σ, the discriminant are

$$\boxed{g_i(x) = \mu_i^T \Sigma^{-1} x - \frac{1}{2} \mu_i^T \Sigma^{-1} \mu_i + \log P(C_i)}$$

$\Sigma = I \Rightarrow \Sigma^{-1} = I$ ← Identity

$$\boxed{\because \quad \Sigma \Sigma^{-1} = I \leftarrow \text{General Identity } A A^{-1} = I}$$

$$\Rightarrow g_i(x) = \mu_i^T x - \frac{1}{2} \mu_i^T \mu_i + \log P(C_i)$$

$$g_1(x) = \mu_1^T x - \frac{1}{2} \mu_1^T \mu_1 + \log P(C_1)$$

$$= \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \log\left(\frac{1}{2}\right)$$

$$= x_1 + x_2 - \frac{1}{2}(2) + \log\left(\frac{1}{2}\right)$$

$$= x_1 + x_2 + \log(0.5) - 1$$

$$g_2(x) = \mu_2^T x - \frac{1}{2} \mu_2^T \mu_2 + \log P(C_2)$$

$$= \begin{bmatrix} 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 3 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} + \log(0.5)$$

$$= 3x_1 + x_2 - 5 + \log(0.5)$$

$$= 3x_1 + x_2 + \log(0.5) - 5$$

(b) $g(x_1) = g(x_2)$ for boundary

$$\Rightarrow x_1 + \cancel{x_2} + \cancel{\log(0.5)} - 1 = 3x_1 + \cancel{x_2} + \cancel{\log(0.5)} - 5$$

$$2x_1 = 4$$

$$x_1 = 2$$

(c) $x = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$

$$g_1(x) = 2 - 1 + \log\left(\tfrac{1}{2}\right)$$
$$= 1 + \log\left(\tfrac{1}{2}\right)$$

$$g_2(x) = 6 - 5 + \log\left(\tfrac{1}{2}\right)$$
$$= 1 + \log\left(\tfrac{1}{2}\right)$$

This point lies on the boundary.

Q10

$$P(C_1 | x) = \sigma(ax + b)$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$a = 2, \; b = -2$

$P(C_1 | x) > \frac{1}{2} \Rightarrow C_1$

else C_2

(a) $P(C_1 | x) = \sigma(ax + b) = \sigma(2x - 2)$

$$= \frac{1}{1 + e^{-(2x-2)}}$$

(b) $P(C_1|x) = \frac{1}{2}$

$$\Rightarrow \frac{1}{1+e^{-(2x-2)}} = \frac{1}{2}$$

$$\Rightarrow 2 = 1 + e^{-(2x-2)}$$

$$\Rightarrow e^{-(2x-2)} = 1$$

$$\Rightarrow 2x - 2 = 0$$

$$\Rightarrow x = 1$$

$$\left[\begin{array}{l} \because e^{-(2x-2)} = 1 \\ -(2x-2)\ln e = \ln(1) \\ \Rightarrow -(2x-2) = 0 \\ \therefore x = 1 \end{array} \right]$$

(c)

$x = 0$	$x = 1$	$x = 2$
$\frac{1}{1+e^{+2}}$	$\frac{1}{1+e^{0}}$	$\frac{1}{1+e^{-2}}$
≈ 0.12	$\frac{1}{2}$	≈ 0.88
$\Rightarrow C_1$	boundary	C_1

Q11

$$P(C_k|x) = \frac{e^{a_k}}{\sum_{j=1}^{3} e^{a_j}}$$

$a_k = w_k^T x + w_0$

$a_1 = 1 \quad a_2 = 0 \quad a_3 = -1$

(a) $P(C_1|x) = \dfrac{e^{a_1}}{e^{a_1}+e^{a_2}+e^{a_3}}$

$P(C_2|x) = \dfrac{e^{a_2}}{e^{a_1}+e^{a_2}+e^{a_3}}$

$P(C_3|x) = \dfrac{e^{a_3}}{e^{a_1}+e^{a_2}+e^{a_3}}$

(b)

$$\boxed{\begin{array}{l} P(A|B) \leftarrow \text{Posterior} \\ P(B|A) \leftarrow \text{likelihood} \\ P(B) \leftarrow \text{Prior} \end{array}}$$

So for this question we need to find $P(C_1|x)$, $P(C_2|x)$ & $P(C_3|x)$

$P(C_1|x) = \dfrac{e^1}{e^1+e^0+e^{-1}} = \underline{0.665} \in x$

$P(C_2|x) = \dfrac{e^0}{e^1+e^0+e^{-1}} = 0.25$

$P(C_3|x) = \dfrac{e^{-1}}{e^1+e^0+e^{-1}} = 0.09$

(c) $P(C_1|x) + P(C_2|x) + P(C_3|x) = \dfrac{e^1+e^0+e^{-1}}{e^1+e^0+e^{-1}} = 1$

Q4

$c_1 \rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ $\quad x_1 = -1$, $x_2 = 0$, $x_3 = 1$

$c_2 \rightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ $\quad x_1 \in c_1$, $x_2 \in c_2$, $x_3 \in c_2$

$c_3 \rightarrow \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ $\quad y(x) = \tilde{w}^T \tilde{x} \quad \tilde{x} = \begin{bmatrix} 1 \\ x \end{bmatrix}$

(a) $T = [\, c_1 \;\; c_2 \;\; c_3 \,] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(b) $\tilde{x}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \tilde{x}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \tilde{x}_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$\tilde{X} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \left(\text{i.e} \begin{bmatrix} \tilde{x}_1^T \\ \tilde{x}_2^T \\ \tilde{x}_3^T \end{bmatrix} \right)$$

(c) $\tilde{w} = (\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T T$

$$\tilde{X}^T \tilde{X} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

$$(\tilde{X}^T \tilde{X})^{-1} = \begin{bmatrix} 1/3 & 0 \\ 0 & 1/2 \end{bmatrix} \quad \boxed{\because A A^{-1} = I}$$

$(\hat{X}^T\hat{X})^{-1}\hat{X}T$ we do this first to ensure dimension match $(\hat{X}^T\hat{X})^{-1}$ currently we cannot multiply $(\hat{X}^T\hat{X})^{-1}\hat{X}$ due to dimension mismatch

$$\hat{X}T = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

$$(\hat{X}^T\hat{X})^{-1}(\hat{X}T) = \begin{bmatrix} 1/3 & 0 \\ 0 & 1/2 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ -1/2 & 0 & 1/2 \end{bmatrix}$$

(d) $$y(x) = \begin{bmatrix} 1/3 & -1/2 \\ 1/3 & 0 \\ 1/3 & 1/2 \end{bmatrix}\begin{bmatrix} 1 \\ x \end{bmatrix}$$

$$= \begin{bmatrix} 1/3 - \frac{1}{2}x \\ 1/3 \\ 1/3 + 1/2\,x \end{bmatrix} \begin{matrix} \leftarrow y_1(x) \\ \leftarrow y_2(x) \\ \leftarrow y_3(x) \end{matrix}$$

(e) for $x = -1$

$y_1(-1) = 5/6$ C_1 $\quad y_2(-1) = 1/3 \quad y_3(-1) = -1/6$

for $x = 1$ $\quad y_1(1) = -1/6 \quad y_2(1) = 1/3 \quad y_3(1) = 5/6$ C_3

for $x = 0$ $\quad y_1(0) = 1/3 = y_2(0) = y_3(0)$ boundary

Q3

(a) target vector for ∴ this is vector for C_1

& (b)

$$x_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$x_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad x_4 \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

(c) $y = (0.1,\ 0.2,\ \underline{0.6},\ 0.1)^T$

↑ higher

so $(0, 0, 1, 0)$ or C_3

(d) $y' = (0, 1, 0, 0)^T$

$\therefore x' \in C_2$

Q2

(a) $C_1 \rightarrow y_1(x) > y_2(x)$ & $y_1(x) > y_3(x)$

$C_2 \rightarrow y_2(x) > y_1(x)$ & $y_2(x) > y_3(x)$

$C_3 \rightarrow y_3(x) > y_1(x)$ & $y_3(x) > y_2(x)$

(b) $y_1(x) = y_2(x)$

or $x_1 = x_2$

(c) $y_1(x) = y_3(x)$

or $x_1 = 0$

$y_2(x) = y_3(x)$

$\rightarrow x_2 = 0$

(d) $x_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ $y_1(x) > y_2(x)$

$\therefore C_1$

$x_2 \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ $y_2(x) > y_1(x)$ & C_2

$x_3 = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$ $y_3(x) > y_1(x)$ & $y_3(x) > y_2(x)$ & C_3